

Chapter 1H: Valvular Heart Disease and Auscultation

Medvale Internal Medicine - Cardiovascular Medicine

This original chapter is designed for custom graph-RAG use. The uploaded valvular heart disease supplement was used as a coverage checklist for mitral stenosis, aortic stenosis, aortic regurgitation, mitral regurgitation, tricuspid regurgitation, mitral valve prolapse, rheumatic heart disease, infective endocarditis, nonbacterial thrombotic endocarditis, and Libman-Sacks endocarditis. Current ACC/AHA valvular guidance and AHA/ADA infective endocarditis prophylaxis guidance were used to update high-yield thresholds and management framing.

Why this matters clinically

Valvular heart disease is one of the best examples of bedside physiology. A murmur is not just a sound; it is a pressure gradient, a flow abnormality, or a structural failure that becomes audible. A stenotic valve blocks forward flow, increases pressure upstream, and eventually causes chamber hypertrophy or failure. A regurgitant valve leaks backward, imposes a volume load, and eventually causes chamber dilation, pulmonary congestion, atrial fibrillation, or systemic embolization depending on the valve involved.

The clinical task is to connect the sound, the chamber response, the symptoms, and the timing of intervention. Aortic stenosis can remain silent for years but becomes dangerous when angina, syncope, or heart failure appears. Mitral stenosis may present as exertional dyspnea, hemoptysis, atrial fibrillation, or embolic stroke because a chronically pressured left atrium becomes enlarged and thrombogenic. Mitral regurgitation and aortic regurgitation may be well tolerated when chronic, but acute regurgitation can cause pulmonary edema and shock because the receiving chamber has no time to adapt.

For graph-RAG, the most important associations are valve lesion -> pressure or volume load -> chamber remodeling -> symptoms -> murmur -> echocardiographic severity -> intervention threshold. This chapter builds those associations in prose, with numeric anchors and exam clues embedded throughout.

Core principle

Valve disease is not managed by murmur loudness alone. Management depends on symptoms, ventricular size and function, pulmonary pressures, rhythm complications, embolic risk, and echocardiographic severity.

Heart sounds and murmur vocabulary

S1 reflects closure of the mitral and tricuspid valves at the start of systole. S2 reflects closure of the aortic and pulmonic valves at the end of systole. A systolic murmur occurs between S1 and S2; a diastolic murmur occurs after S2 before the next S1. As a rule, diastolic murmurs are pathologic until proven otherwise because normal diastolic flow should not create loud turbulence.

An S3 occurs early in diastole during rapid ventricular filling. In a young healthy patient it can be physiologic, but in adults it often suggests volume overload or systolic heart failure. An S4 occurs late in diastole when the atrium contracts against a stiff ventricle; it is associated with hypertrophy, ischemia, hypertension, and diastolic dysfunction. An opening snap is a high-frequency sound after S2 caused by sudden tensing of a stenotic mitral valve. In mitral stenosis, the shorter the interval between A2 and the opening snap, the more severe the stenosis because higher left atrial pressure forces the valve open earlier.

Sound or murmur	Timing	High-yield association
S3	Early diastole	Volume overload, HFrEF, dilated ventricle; often heard at apex with bell
S4	Late diastole	Stiff ventricle from LVH, ischemia, restrictive disease; absent in atrial fibrillation
Opening snap	After S2	Mitral stenosis; shorter A2-OS interval suggests more severe MS
Mid-systolic click	Mid to late systole	Mitral valve prolapse
Diastolic decrescendo murmur	Early diastole	Aortic regurgitation
Holosystolic murmur	S1 through S2	Mitral regurgitation, tricuspid regurgitation, VSD

Maneuvers: turning physiology into sound

Murmur maneuvers work by changing preload, afterload, chamber size, or venous return. Inspiration increases right-sided venous return, so right-sided murmurs such as tricuspid regurgitation usually increase with inspiration. Handgrip increases systemic vascular resistance and afterload; it tends to increase regurgitant murmurs such as mitral regurgitation and aortic regurgitation because blood has more resistance to forward ejection and more tendency to leak backward. Standing and Valsalva reduce venous return and left ventricular size. Most murmurs soften with decreased preload, but hypertrophic cardiomyopathy and mitral valve prolapse become more prominent because a smaller LV worsens dynamic obstruction or allows the prolapsing leaflet to click earlier. Squatting increases venous return and afterload; it usually softens HCM and delays MVP click.

Maneuver	Physiologic effect	Murmurs that increase or key clue
Inspiration	Increases right-sided venous return	Tricuspid regurgitation, pulmonic stenosis
Handgrip	Increases afterload/SVR	Mitral regurgitation and aortic regurgitation increase; HCM often decreases
Standing/Valsalva	Decreases preload and LV size	HCM louder; MVP click/murmur occurs earlier
Squatting	Increases preload and afterload	Most murmurs louder; HCM softer; MVP click delayed

Numbers to know: valve disease

Topic	High-yield numeric anchor
Normal mitral valve area	About 4-5 cm ²
Mitral stenosis symptoms often appear	Usually when mitral valve area falls to about 1.5 cm ² or less
Severe aortic stenosis	Aortic valve area <1.0 cm ² , Vmax ≥4.0 m/s, or mean gradient ≥40 mmHg
Aortic stenosis prognosis after symptoms	Classic teaching: angina ~3 years, syncope ~2 years, heart failure ~1.5 years without valve replacement
Severe chronic aortic regurgitation intervention signal	Symptoms or LV systolic dysfunction; guideline threshold commonly includes LVEF ≤55% for severe AR
Severe primary mitral regurgitation LV dysfunction threshold	Surgery recommended when LVEF ≤60% or LVESD ≥40 mm in appropriate severe primary MR
Endocarditis blood cultures	Draw 3 sets before antibiotics when possible
Duke criteria diagnosis	Definite IE classically: 2 major, or 1 major + 3 minor, or 5 minor criteria
Acute endocarditis tempo	Can be fatal in <6 weeks if untreated; often Staphylococcus aureus
Subacute endocarditis tempo	Often indolent >6 weeks; classically viridans streptococci or enterococci

Mitral stenosis

Mitral stenosis is usually a delayed consequence of rheumatic heart disease. The initial streptococcal infection is in the throat, but immune cross-reactivity injures the valve, producing commissural fusion, leaflet thickening, calcification, and a narrow fish-mouth mitral orifice. The patient may not remember the childhood or adolescent episode of rheumatic fever, so the absence of that history does not exclude the diagnosis.

The hemodynamic problem in mitral stenosis is obstruction between the left atrium and left ventricle during diastole. Blood has difficulty crossing the mitral valve, so left atrial pressure rises. The left atrium enlarges, pulmonary venous pressure increases, and pulmonary congestion causes exertional dyspnea, orthopnea, and paroxysmal nocturnal dyspnea. With long-standing disease, pulmonary hypertension places pressure overload on the right ventricle, producing right ventricular failure with hepatomegaly, ascites, edema, JVD, and a loud P2.

The normal mitral valve area is roughly 4-5 cm². Many patients become symptomatic when the valve area falls to about 1.5 cm². Tachycardia worsens symptoms because a faster heart rate shortens diastole, leaving even less time for left atrial emptying. Pregnancy, exercise, fever, anemia, thyrotoxicosis, and atrial fibrillation can all unmask or worsen mitral stenosis by increasing heart rate, blood volume, or flow across the narrowed valve.

Atrial fibrillation is a major complication of mitral stenosis because chronic left atrial enlargement disrupts atrial electrical architecture. AF removes the atrial kick and makes ventricular filling more irregular, worsening symptoms. It also promotes thrombus formation in the left atrium and left atrial appendage, creating embolic stroke risk.

Mitral stenosis feature	Association
Murmur	Opening snap followed by low-pitched diastolic rumble at the apex
Best heard	Bell of stethoscope, left lateral decubitus position
Severity clue	Shorter A2-opening snap interval means more severe stenosis
Pulmonary clue	Dyspnea, orthopnea, PND, hemoptysis from pulmonary venous hypertension
Rhythm clue	AF due to left atrial enlargement; anticoagulate when AF develops
Diagnosis	Echocardiography is the key confirmatory test
Treatment	Diuretics for congestion, beta-blockers/rate control to lengthen diastole, balloon valvotomy or surgery for severe disease

Aortic stenosis

Aortic stenosis is obstruction at the left ventricular outflow valve. The left ventricle must generate higher pressure to eject blood across a narrowed aortic valve, so it develops concentric hypertrophy. At first, hypertrophy preserves systolic performance. Over time, the ventricle becomes stiff, diastolic filling worsens, myocardial oxygen demand rises, and coronary reserve falls. Eventually the ventricle can no longer compensate, leading to angina, syncope, heart failure, and death if the valve is not replaced.

The common causes depend on age. A congenitally bicuspid aortic valve can calcify earlier in adulthood. Degenerative calcific disease is common in older adults. Rheumatic disease can also involve the aortic valve, often with other valve lesions. Severe aortic stenosis is often defined by an aortic valve area less than 1.0 cm², a peak aortic jet velocity at least 4.0 m/s, or a mean gradient at least 40 mmHg, although low-flow and low-gradient states require more nuanced interpretation.

The classic symptomatic triad is angina, syncope, and heart failure. Angina can occur even without epicardial CAD because the hypertrophied LV has high oxygen demand and impaired subendocardial perfusion. Syncope is often exertional because the stenotic valve prevents cardiac output from rising appropriately during activity. Heart failure reflects decompensation of the hypertrophied or dilated ventricle. Older exam teaching links symptom onset to poor survival without intervention: angina about 3 years, syncope about 2 years, and heart failure about 1.5 years.

The murmur of aortic stenosis is a harsh crescendo-decrescendo systolic ejection murmur best heard at the right upper sternal border and radiating to the carotids. The carotid upstroke may be delayed and diminished, called parvus et tardus. Severe AS may produce a soft or single S2 because the calcified aortic valve closes poorly or late.

Exam trap: symptomatic severe AS

A symptomatic patient with severe aortic stenosis generally needs valve replacement evaluation, not reassurance. Exercise stress testing is avoided in symptomatic severe AS because of the risk of hemodynamic collapse.

Aortic stenosis question	Answer
Murmur	Harsh crescendo-decrescendo systolic murmur at RUSB radiating to carotids
Pulse finding	Parvus et tardus: weak, delayed carotid upstroke
Severe echo anchors	AVA <1.0 cm ² , Vmax ≥4.0 m/s, mean gradient ≥40 mmHg
Symptoms	Angina, syncope, dyspnea/heart failure
Main treatment	Aortic valve replacement: SAVR or TAVR depending on age, anatomy, risk, and shared decision-making

Aortic regurgitation

Aortic regurgitation is failure of the aortic valve to close during diastole. Blood that should remain in the aorta leaks backward into the left ventricle. Chronic AR creates a volume-loaded LV that adapts by eccentric dilation and hypertrophy. For years, the ventricle may maintain forward stroke volume by enlarging. Eventually compensatory dilation fails, LV end-diastolic pressure rises, pulmonary congestion develops, and systolic function declines.

The causes can be divided into leaflet disease and aortic root disease. Leaflet causes include bicuspid aortic valve, rheumatic disease, infective endocarditis, and trauma. Aortic root causes include aortic dissection, Marfan syndrome, syphilitic aortitis, ankylosing spondylitis, giant cell arteritis, and chronic hypertension. Acute AR, such as from dissection, endocarditis, or trauma, is an emergency because the LV has no time to dilate; diastolic pressure rises suddenly, pulmonary edema and shock can occur, and emergent surgery may be required.

The murmur of chronic AR is an early diastolic decrescendo murmur best heard along the left sternal border with the patient sitting forward in expiration. Wide pulse pressure is a major bedside clue because systolic pressure rises from increased stroke volume while diastolic pressure falls as blood runs back into the LV. Bounding pulses, head bobbing, pistol-shot femoral sounds, and capillary pulsations are classic peripheral signs, but they are not required for diagnosis. The Austin Flint murmur is a low-pitched diastolic rumble at the apex caused by regurgitant flow interfering with mitral valve opening, mimicking mitral stenosis.

Echocardiography evaluates severity, LV size, LV systolic function, aortic root anatomy, and associated valve disease. In asymptomatic severe chronic AR, intervention is considered before irreversible LV dysfunction develops; current guideline thresholds include symptoms or LV systolic dysfunction, and LVEF $\leq 55\%$ is an important severe AR trigger in modern ACC/AHA staging.

Aortic regurgitation feature	Association
Murmur	Early diastolic decrescendo murmur at left sternal border
Pulse pressure	Wide: high systolic, low diastolic
Chronic physiology	LV volume overload -> eccentric dilation -> eventual systolic dysfunction
Acute physiology	No time for LV adaptation -> pulmonary edema/shock -> surgical emergency
Austin Flint murmur	Apical diastolic rumble from regurgitant jet functionally impeding mitral opening
Treatment logic	Afterload reduction in selected chronic symptomatic/severe disease; valve replacement for symptoms or significant LV dysfunction

Mitral regurgitation

Mitral regurgitation is backward flow from the left ventricle into the left atrium during systole. The physiology depends on whether the regurgitation is acute or chronic. In acute MR, the left atrium is normal-sized and noncompliant. Sudden regurgitant flow sharply raises left atrial and pulmonary venous pressure, causing flash pulmonary edema and hypotension because forward stroke volume falls. Acute MR can occur after papillary muscle rupture from MI, infective endocarditis, chordae rupture, or trauma.

In chronic MR, the left atrium and left ventricle gradually dilate to accommodate the regurgitant volume. This adaptation allows symptoms to be delayed, but it is not benign. Chronic volume overload eventually causes LV dilation, LV systolic dysfunction, pulmonary hypertension, atrial fibrillation, and heart failure. Because the LV ejects both forward into the aorta and backward into the low-pressure LA, ejection fraction can appear deceptively preserved until disease is advanced. In severe primary MR, an LVEF of 60% is already a warning threshold rather than reassuring normal function, and LV end-systolic dimension ≥ 40 mm is another key trigger for surgery in appropriate patients.

The murmur is a holosystolic murmur best heard at the apex and often radiating to the axilla or back. The radiation direction can reflect which leaflet is diseased. Associated findings include a soft S1, S3 from volume overload, laterally displaced PMI, pulmonary edema in acute MR, and AF in chronic MR. Echocardiography defines the mechanism, chamber size, regurgitant severity, pulmonary pressures, and feasibility of repair. Durable surgical repair is preferred when possible for primary degenerative MR; transcatheter edge-to-edge repair is considered in selected high-risk patients or specific secondary MR contexts.

MR distinction	Acute MR	Chronic MR
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LA response	Normal size, noncompliant, high pressure	Dilated, more compliant initially
Presentation	Pulmonary edema, hypotension, shock	Dyspnea, AF, progressive HF
Common causes	Papillary muscle rupture, endocarditis, chordae rupture	MVP, rheumatic disease, cardiomyopathy, annular dilation
Murmur	May be softer if pressures equalize quickly	Holosystolic apical murmur radiating to axilla/back
Management	Urgent stabilization and surgery often needed	Serial echo; repair/replacement before irreversible LV dysfunction

Tricuspid regurgitation

Tricuspid regurgitation is systolic backflow from the right ventricle into the right atrium. Mild TR is common on echocardiography and may be physiologic, but clinically significant TR reflects right-sided volume overload, annular dilation, pulmonary hypertension, endocarditis, carcinoid disease, congenital leaflet abnormalities, or device lead interference. The most common mechanism is functional TR from RV dilation, often secondary to left-sided heart failure or pulmonary hypertension.

The symptoms are those of right-sided congestion: peripheral edema, ascites, hepatomegaly, abdominal fullness, and JVD. The murmur is a blowing holosystolic murmur at the lower left sternal border that increases with inspiration, known as Carvallo sign, because inspiration increases venous return to the right heart. Pulsatile liver and prominent v waves in the jugular venous pulse can occur with severe TR. Diagnosis is made by echocardiography, which quantifies regurgitation, evaluates leaflet anatomy, estimates pulmonary pressures, and assesses RV/RA size and function.

Treatment begins with the cause. Diuretics reduce venous congestion. Pulmonary hypertension, left-sided HF, endocarditis, or device-related problems should be addressed when present. Severe symptomatic TR may require surgical or transcatheter intervention, but advanced pulmonary hypertension and RV dysfunction complicate decision-making.

Mitral valve prolapse

Mitral valve prolapse is excessive or redundant mitral leaflet tissue that bows into the left atrium during systole. It is commonly linked to myxomatous degeneration and connective tissue disorders such as Marfan syndrome, Ehlers-Danlos syndrome, and osteogenesis imperfecta. Many patients are asymptomatic, and the condition is often benign.

The classic auscultatory finding is a mid-systolic click followed by a late systolic murmur. Maneuvers that reduce LV volume, such as standing and Valsalva, allow the valve to prolapse earlier, so the click occurs earlier and the murmur lengthens. Squatting increases LV volume, delays prolapse, and makes the click occur later. Palpitations and atypical chest pain may occur, but sudden death is rare. Echocardiography confirms leaflet prolapse and identifies clinically important MR. Reassurance is appropriate for many asymptomatic patients; beta-blockers may help selected patients with palpitations or chest discomfort.

Condition	Murmur/sound	Maneuver clue
Mitral valve prolapse	Mid-systolic click +/- late systolic murmur	Standing/Valsalva earlier click; squatting later click
Tricuspid regurgitation	Holosystolic murmur at LLSB	Louder with inspiration
Mitral regurgitation	Holosystolic apical murmur	Louder with handgrip
Aortic regurgitation	Early diastolic decrescendo murmur	Louder with handgrip; wide pulse pressure
Aortic stenosis	Crescendo-decrescendo systolic murmur at RUSB	Radiates to carotids; parvus et tardus
Mitral stenosis	Opening snap + diastolic rumble	Best in left lateral decubitus; shorter A2-OS interval

Rheumatic heart disease

Rheumatic heart disease is chronic valvular damage that follows acute rheumatic fever, an immune-mediated sequela of group A streptococcal pharyngitis. Acute rheumatic fever can produce migratory polyarthritis, carditis, chorea, erythema marginatum, and subcutaneous nodules. Chronic rheumatic heart disease most often causes mitral stenosis, but it can also cause mitral regurgitation, aortic regurgitation, aortic stenosis, or multivalve disease.

Diagnosis of acute rheumatic fever uses major and minor Jones criteria plus evidence of preceding streptococcal infection. Major criteria include migratory polyarthritis, carditis, chorea, erythema marginatum, and subcutaneous nodules. Minor criteria include fever, elevated ESR/CRP, arthralgias, prolonged PR interval, and prior rheumatic fever. Treatment includes eradication of streptococcal infection, anti-inflammatory therapy, monitoring of carditis, and secondary prophylaxis to prevent recurrent rheumatic fever.

Infective endocarditis

Infective endocarditis is infection of the endocardial surface, usually involving a valve. It creates vegetations composed of microorganisms, platelets, fibrin, and inflammatory debris. These vegetations can destroy valves, seed the bloodstream, embolize to distant organs, and produce immune phenomena. The disease can present acutely with sepsis and valve destruction or subacutely with low-grade fever, malaise, weight loss, embolic events, and a new murmur.

Organisms depend on the clinical setting. *Staphylococcus aureus* is a major cause of acute endocarditis, endocarditis in injection drug use, and healthcare-associated disease. Viridans streptococci classically follow dental manipulation or poor dentition and often cause subacute disease on abnormal native valves. Enterococci may follow genitourinary or gastrointestinal sources. *Staphylococcus epidermidis* is important in prosthetic valve infections, especially early after surgery. HACEK organisms are slow-growing gram-negative oral flora associated with culture-negative or indolent endocarditis. Fungal endocarditis is less common but important in prosthetic valves, immunocompromise, and injection drug use.

The key diagnostic action is to obtain three sets of blood cultures before antibiotics when the patient is stable enough to wait. Treatment is prolonged bactericidal intravenous antibiotics tailored to organism and susceptibility. Echocardiography evaluates vegetations, abscesses, valve destruction, and prosthetic valve involvement. Transthoracic echo is often the initial test, while transesophageal echo has better sensitivity for small vegetations, prosthetic valves, abscess, and complicated disease.

Endocarditis clue	Meaning
Fever + new murmur	Always consider infective endocarditis
Janeway lesions	Painless palms/soles lesions from septic emboli/microabscesses
Osler nodes	Painful finger/toe nodules from immune complex deposition
Roth spots	Retinal hemorrhages with pale centers
Splinter hemorrhages	Linear nail-bed hemorrhages
Stroke or renal infarct	Systemic embolization from left-sided vegetations
AV block	Possible paravalvular abscess involving conduction system
Right-sided IE	Often injection drug use; septic pulmonary emboli

Duke criteria framework

Definite infective endocarditis is classically diagnosed by 2 major criteria, or 1 major plus 3 minor criteria, or 5 minor criteria. Major criteria include typical sustained bacteremia and echocardiographic evidence such as vegetation, abscess, prosthetic valve dehiscence, or new valvular regurgitation.

When to involve surgery in endocarditis

Surgical consultation is urgent when endocarditis causes acute heart failure from valve dysfunction, persistent bacteremia despite appropriate antibiotics, abscess or heart block, highly resistant organisms, fungal infection, recurrent embolization, large mobile vegetations with embolic risk, or prosthetic valve infection. The key concept is that antibiotics alone may sterilize blood but cannot always fix a destroyed valve, drain an abscess, or remove infected prosthetic material.

Endocarditis prophylaxis

Endocarditis prophylaxis is narrower than older teaching. Current AHA/ADA framing limits prophylaxis to patients at highest risk of adverse outcomes from infective endocarditis and only for specific procedures. High-risk cardiac conditions include prosthetic cardiac valves or prosthetic material used for valve repair, prior infective endocarditis, selected congenital heart disease categories, and cardiac transplant with valvulopathy. Dental procedures that manipulate gingival tissue, the periapical region of teeth, or perforate oral mucosa are the classic qualifying procedures. Routine GI or GU procedures do not receive prophylaxis solely to prevent endocarditis. Native mitral valve prolapse or native mitral stenosis alone is not a prophylaxis indication.

Prophylaxis usually yes when BOTH high-risk heart condition and qualifying procedure	Prophylaxis usually no
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Prosthetic valve or prosthetic valve repair material + invasive dental procedure	Native MVP, native MS, or routine low-risk murmurs alone
Prior infective endocarditis + invasive dental procedure	Routine GI endoscopy or cystoscopy solely for IE prevention
Certain unrepaired/recently repaired congenital heart disease + qualifying dental procedure	Most nondental procedures in the absence of active infection
Cardiac transplant with valvulopathy + qualifying dental procedure	Dental radiographs, local anesthetic injections through noninfected tissue, shedding deciduous teeth

Noninfective vegetations

Not all valve vegetations are infectious. Nonbacterial thrombotic endocarditis, also called marantic endocarditis, consists of sterile fibrin-platelet deposits along valve closure lines. It is associated with malignancy, severe systemic illness, and hypercoagulability. The vegetations are friable and can embolize to the brain, spleen, kidneys, or extremities, but blood cultures are negative because there is no infection. Management targets the underlying disease and hypercoagulable state.

Libman-Sacks endocarditis is a sterile verrucous endocarditis associated with systemic lupus erythematosus and antiphospholipid syndrome. It classically affects the mitral or aortic valve and may form small warty vegetations on both sides of the valve leaflets. It can cause regurgitant murmurs and embolic events. Treatment focuses on the underlying autoimmune disease and anticoagulation when thromboembolic risk or antiphospholipid syndrome is present.

High-yield integrated comparison

Lesion	Main physiology	Classic symptom pattern	Classic sound
Mitral stenosis	LA pressure overload -> pulmonary venous HTN	Dyspnea, hemoptysis, AF, emboli	Opening snap + low diastolic rumble
Aortic stenosis	LV pressure overload -> concentric LVH	Angina, syncope, HF	Harsh systolic crescendo-decrescendo at RUSB to carotids
Aortic regurgitation	LV volume overload; acute AR causes shock	Wide pulse pressure, dyspnea; acute pulmonary edema	Early diastolic decrescendo murmur
Mitral regurgitation	LA/LV volume overload	Dyspnea, AF; acute MR flash pulmonary edema	Holosystolic apical murmur radiating to axilla
Tricuspid regurgitation	RA/RV volume overload, systemic venous congestion	Edema, ascites, hepatomegaly, JVD	Holosystolic LLSB murmur louder with inspiration
MVP	Leaflet prolapse into LA during systole	Often asymptomatic; palpitations/atypical pain	Mid-systolic click +/- late systolic murmur

Common pitfalls

One pitfall is assuming a quiet murmur means mild disease. Acute MR can produce a relatively soft murmur because left atrial pressure rises rapidly and the pressure gradient between LV and LA falls. Another pitfall is treating symptomatic severe AS medically for too long; medications may palliate comorbid hypertension or congestion, but valve replacement is definitive therapy when symptoms and severe stenosis are present. A third pitfall is missing atrial fibrillation in mitral stenosis, where anticoagulation and rate control become central because left atrial enlargement promotes thrombus.

A fourth pitfall is giving endocarditis prophylaxis too broadly. Modern prophylaxis is reserved for high-risk cardiac conditions and qualifying procedures, especially invasive dental procedures involving gingival or periapical manipulation or mucosal perforation. A fifth pitfall is forgetting noninfective vegetations when blood cultures are negative and embolic phenomena occur in a patient with malignancy, lupus, or antiphospholipid syndrome.

Practice questions

Question 1. A 48-year-old immigrant has progressive dyspnea and new atrial fibrillation. Exam reveals an opening snap followed by a low-pitched diastolic rumble at the apex. What is the likely diagnosis?

Answer: Mitral stenosis from chronic rheumatic heart disease. The opening snap and diastolic rumble are classic, and AF develops from left atrial enlargement.

Question 2. A 76-year-old has exertional syncope, delayed carotid upstrokes, and a harsh systolic murmur at the right upper sternal border radiating to the carotids. Echo shows Vmax 4.3 m/s. What is the key management direction?

Answer: Symptomatic severe aortic stenosis; evaluate for aortic valve replacement with SAVR or TAVR depending on risk, age, anatomy, and shared decision-making.

Question 3. A patient develops acute pulmonary edema and hypotension after an inferior MI. A new holosystolic murmur is heard at the apex. What complication is most concerning?

Answer: Acute mitral regurgitation from papillary muscle rupture or ischemic dysfunction. Acute MR can produce flash pulmonary edema and shock.

Question 4. A patient with a prosthetic aortic valve has fever and a new murmur. How many blood culture sets should be obtained before antibiotics if stable?

Answer: Three sets of blood cultures should be drawn before antibiotics when clinically feasible.

Question 5. A young woman with SLE has embolic strokes and sterile vegetations on both sides of valve leaflets. What is the diagnosis?

Answer: Libman-Sacks endocarditis, a noninfective verrucous endocarditis associated with SLE/antiphospholipid syndrome.

Chapter summary

Valvular disease is best learned by connecting physiology to sound. Mitral stenosis causes left atrial pressure overload, pulmonary congestion, atrial fibrillation, and an opening snap with a diastolic rumble. Aortic stenosis causes LV pressure overload, concentric hypertrophy, angina, syncope, heart failure, and a harsh systolic murmur radiating to the carotids; severe AS is anchored by AVA $<1.0 \text{ cm}^2$, $V_{\text{max}} \geq 4.0 \text{ m/s}$, or mean gradient $\geq 40 \text{ mmHg}$. Aortic regurgitation creates LV volume overload, wide pulse pressure, and an early diastolic decrescendo murmur; acute AR is a surgical emergency. Mitral regurgitation creates LA/LV volume overload, an apical holosystolic murmur, and risk for AF and HF; severe primary MR requires attention before EF falls below the deceptively high threshold of about 60%. Tricuspid regurgitation causes systemic venous congestion and a holosystolic LLSB murmur louder with inspiration. Endocarditis adds the infection framework: fever, new murmur, emboli, three blood culture sets, Duke criteria, echo, prolonged bactericidal antibiotics, and surgery when valve destruction or uncontrolled infection occurs.

Selected source notes

1. Uploaded cardiology5.pdf, pages 1-20: used as coverage checklist for valve lesions, rheumatic heart disease, infective endocarditis, prophylaxis, NBTE, and Libman-Sacks endocarditis. 2. 2020 ACC/AHA valvular heart disease guideline and ACC summary materials: used for contemporary severity/intervention thresholds including AS, AR, and MR. 3. American Heart Association/American Dental Association infective endocarditis prophylaxis materials: used for modern prophylaxis framing.